— SOLUTIONS —

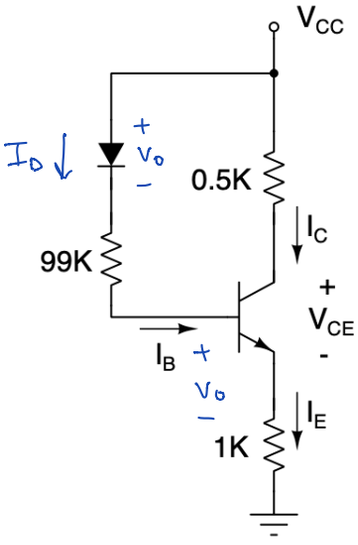
EEE211 Quiz#3, Fall 2022 21.11.2022

Name: **Student ID:**

Duration: 20 minutes

For the circuit shown below. V_{CC} = 20 V, V_0 = 0.7 V for both BJT and diode, V_{CEsat} = 0.2 V, β = 100.

a) **(50 pts)** Find the currents I_B, I_C, and I_E.
b) **(25 pts)** Find the voltage V_{CE}.
c) **(25 pts)** Specify and verify the states of the diode and the BJT.



a) Assume diode is ON, BJT is in ACT state

KVL: $V_{CC} = V_o + 99K \cdot I_B + V_o + 1K \cdot \underbrace{I_E}_{(\beta+1) I_B}$

$$V_{CC} - 2V_o = (99K + 101K) I_B$$

$$I_B = \frac{20 - 1.4}{200K} = \boxed{0.093\,mA}$$

Then,

$$I_C = \beta \cdot I_B = 100 \times 0.093 = \boxed{9.3\,mA}$$

$$I_E = (\beta+1) I_B = 9.3 + 0.093 \approx \boxed{9.39\,mA}$$

b) KVL: $V_{CC} = 0.5K \cdot I_C + V_{CE} + 1K \cdot I_E$

$$20 = 0.5 \times 9.3 + V_{CE} + 9.39$$

$$V_{CE} = 20 - 4.65 - 9.39$$

$$\boxed{V_{CE} = 5.96\,V}$$

c) * Diode is ON. Verify: $I_D = I_B = 0.093\,mA > 0$ ✓ diode is ON

* BJT is in ACT state. Verify:

$I_B = 0.093\,mA > 0$ ✓

$V_{CE\,sat} = 5.96\,V < V_{CEsat}$ ✓

BJT is ACT